
GENERAL RELATIVITY

Tutorial problem set 17, 24.01.2020.

■ PROBLEM 41 Response of matter to gravitational waves.

The equation of motion for the physical degrees of freedom of gravity propagating in Minkowski space-time is

$$\square h_{ij}^{TT} = 0 , \quad (41.1)$$

where h_{ij}^{TT} is a transverse traceless tensor,

$$\partial^i h_{ij}^{TT} = 0 , \quad (h^{TT})^i_i = 0 . \quad (41.2)$$

Equation of motion (41.2) obviously has propagating waves as solutions.

- (a) Assume a plane wave propagating in an arbitrary direction,

$$h_{ij}^{TT} = A_\sigma \epsilon_{ij}^\sigma \cos(\omega t - \vec{k} \cdot \vec{x}) . \quad (41.3)$$

Determine the dispersion relation.

Now assume a wave propagating in the positive z -direction ($\vec{k} = k\hat{z}$) and show that there are two independent polarizations of a gravitational wave, a *plus* one and a *cross* one, corresponding to polarization tensors

$$\epsilon^+ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} , \quad \epsilon^\times = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} . \quad (41.4)$$

Next we want to determine what is the effect this wave has on a group of test particles initially distributed in a circle perpendicular to the propagation direction of the wave. In order to do that we will solve the geodesic deviation equation. Consider the separation vector between two nearby particle trajectories,

$$\frac{D^2}{d\tau^2} S^\mu = R^\mu_{\nu\rho\sigma} \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} S^\sigma . \quad (41.5)$$

- (b) Assume the test particles to move slowly, $dx^\alpha/d\tau \approx (1, 0, 0, 0)$. Show that by expanding the left- and the right-hand side of equation (41.5) to lowest non-vanishing order the geodesic deviation equation becomes

$$\frac{\partial^2}{\partial t^2} S_i = \frac{1}{2} S^j \frac{\partial^2}{\partial t^2} h_{ij}^{TT} . \quad (41.6)$$

- (c) Consider a gravitational wave as in (41.3) with $A_\times = 0$. Show that (41.6) has the following perturbative solution (A_+ serves as a perturbation parameter in equation (41.6))

$$S^1 = \left(1 + \frac{A_+}{2} \cos(\omega t - kz) \right) S^1(0) , \quad (41.7)$$

$$S^2 = \left(1 - \frac{A_+}{2} \cos(\omega t - kz) \right) S^2(0) . \quad (41.8)$$

Sketch how the particles move assuming they were distributed in a circle at the initial moment.

(d) Repeat the analysis in (c) for $A_+ = 0$.

■ **PROBLEM 42** Gravitational radiation from a binary star system.

Two stars of masses m_1 and m_2 are orbiting each other in non-relativistic circular orbit, separated by a distance R . Since this system loses energy due to gravitational radiation, the separation distance will change in time, $R = R(t)$.

(a) Show that the total (Newtonian) energy of the system is

$$E = -\frac{1}{2} \frac{G_N M \mu}{R}, \quad (42.1)$$

where $M = m_1 + m_2$ and $\mu = \frac{m_1 m_2}{m_1 + m_2}$. Assume that the orbit is a circular one and the separation between the stars is $R = \text{const}$.

(b) Show that the power P_{GW} dissipated by gravitational waves is

$$P_{GW} = \frac{32}{5} \frac{G_N^4 \mu^2 M^3}{R^5}, \quad (42.2)$$

Use the following formula derived in the lectures

$$P_{GW} = \frac{G_N}{5} \langle \partial_0^3 t_{ij} \partial_0^3 t^{ij} \rangle, \quad (42.3)$$

where

$$t_{ij} = \int d^3x T_{00} \left(x_i x_j - \frac{\delta_{ij}}{3} \vec{x}^2 \right). \quad (42.4)$$

(c) The gravitational power is radiated at the expense of orbital energy. Show that the separation between the stars depends on time as

$$R(t) = \left[\frac{256}{5} M^2 \mu (t_0 - t) \right]^{1/4}, \quad (42.5)$$

if we assume the energy loss to be slow enough that the approximation of instantaneously circular orbits holds throughout. Here t_0 is the time when the separation distance goes to zero. Find this time in terms of the masses, and the initial separation distance R_0 .

In 1974 Hulse and Taylor discovered a binary system PSR1913+16, in which both stars are very small, and one of them is a pulsar. The period of the orbit was about eight hours, extremely small by astrophysical standards, and the fact that one of the stars was a pulsar provided a very accurate clock with respect to which the change in the period of the orbit as the system loses energy can be measured. The result was consistent with the prediction from General Relativity for the energy loss through gravitational radiation. Hulse and Taylor were awarded the Nobel prize in 1993 for their work.